Milestone Two: Enhancement One: Software Design/Engineering Narrative

Antoine Boylston

Bachelor of Science in Computer Science, SNHU

CS 499: Computer Science Capstone 2026

Instructor: Alan Spencer, M.S.

March 22, 2026

Narrative

The artifact selected for this enhancement is a **text-based adventure game written in Python**, titled *Dragon King Assassin*. This program was originally created in one of my earliest programming courses as an introduction to control flow, dictionaries, and user input handling. The game allows a player to navigate between rooms, collect items, and attempt to defeat a final boss based on whether sufficient items have been collected. While the original version functioned correctly, it was implemented as a single procedural script, which made it difficult to extend, maintain, and test.

I selected this artifact for inclusion in my ePortfolio because it provides a strong before-and-after demonstration of my growth as a developer. The original version showcases foundational programming skills such as conditionals, loops, and basic data structures. However, the refactored version better represents my current abilities in **software design and engineering practices**. Specifically, I improved the artifact by restructuring it into a modular design consisting of separate components: a game engine, player model, world model, command parser, and a dedicated data module for room definitions. This separation of concerns significantly improves readability, maintainability, and scalability. Additional enhancements included improving input handling through a more robust command parser, replacing broad exception handling with explicit logic checks, and adding a small test file to validate core functionality such as movement, parsing, and inventory management. I also created a README file to document the purpose of the project, its structure, and the improvements made during refactoring.

The enhancements made to this artifact directly support the course outcomes I identified in Module One. Most notably, I met **Outcome 3**, which focuses on designing and evaluating computing solutions using appropriate software engineering practices, by transforming a procedural script into a modular system with clearly defined responsibilities. I also addressed **Outcome 2**, which emphasizes professional communication, through improved code organization and documentation, including the addition of a README and docstrings. Additionally, I supported **Outcome 5** by improving input validation and control flow, reducing the likelihood of unintended behavior and making the program more robust. At this time, I do not need to update my outcome-coverage plans, as the enhancements align well with the goals I originally established.

Reflecting on this process, I learned the importance of structuring code in a way that supports future growth and modification. Initially, my focus as a beginner was simply to make the program work. Through this enhancement, I shifted that mindset toward writing code that is **maintainable, testable, and scalable**. One of the biggest challenges I faced was breaking apart the tightly coupled logic from the original script into separate modules while ensuring the program still functioned correctly. This required me to think more carefully about how different parts of the system interact and where responsibilities should be assigned. I also encountered smaller issues, such as ensuring variables were defined before use and improving the flow of control without relying on broad exception handling. Overall, this process helped reinforce the value of clean architecture and gave me practical experience applying software engineering principles to an existing codebase.

Prompts used (AI Assistance Disclosure)

Generative AI (ChatGPT) was used to assist with refining the clarity, organization, and academic tone. AI assistance was used only for writing support and editing suggestions. All ideas, reflections, and final content decisions were reviewed and approved by the author.

Sample prompt included:

- "Polish this section for clarity and academic tone"
- "Please check this doc for major errors"

All final content was reviewed and edited by the author.